

Generation Interconnection System Impact Study Report

For Queue Project Q16-0661

J K Smith 138 kV
February 25, 2018

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q16-0661. The Interconnection Customer proposes a 107.8 MW battery energy storage facility interconnecting at the J K Smith 138 kV substation in the Kentucky area. The request was received on August 11, 2016. The study evaluated system intact and single contingency conditions using the regional planning case, including 268 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 27 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$9,073,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Douglas Forest / Ellendale 138kV: 133.4% loading (224 MVA) following the outage of the Rancho Carmel / Lanham 138kV; DFAX = 7.2%.

Georgia Pacific / Douglas Forest 138kV: 128.6% loading (227 MVA) following the outage of the Rancho Carmel Substation / Lanham 138kV; DFAX = 7.2%.

For the loss of International Paper / Hanging Rock 500kV L, the Coulee Ave Substation / Sporn 345kV loads to 127.7% of its 1,109 MVA rating. Project DFAX: 6.2%.

The Wellington / Davisville 138kV L is overloaded to 125.1% of its 460 MVA emergency rating for the loss of the Albion Rd / Wixom 138kV Line. The project contributes 5.3% DFAX (5.7 MW) to this overload.

For the loss of Davisville / North Appleton 138kV, the Tule / Santa Rosa Beach 138kV loads to 117.9% of its 176 MVA rating. Project DFAX: 6.5%.

For the loss of Snohomish / Sand Spings 138kV line, the Port Madison / Pinn Rd 138kV L loads to 117.2% of its 247 MVA rating. Project DFAX: 6.1%.

The Southside Wastewater Treatment Plant / Osawatomie 100kV is overloaded to 116.6% of its 143 MVA emergency rating for the loss of the Bluestem / Bodo 230kV. The project contributes 5.5% DFAX (5.9 MW) to this overload.

The Port Madison / Ojo 138kV is overloaded to 116.1% of its 256 MVA emergency rating for the loss of the Sand Springs / Snohomish 138kV. The project contributes 6.1% DFAX (6.6 MW) to this overload.

Guadalupe / Mistersky 138kV: 115.4% loading (230 MVA) following the outage of the Wissota / Berry Street Substation 138kV; DFAX = 9.2%.

For the loss of Osawatomie / Wagner Lake 230kV L, the Bingen / Prager 138kV loads to 114.1% of its 249 MVA rating. Project DFAX: 8.1%.

For the loss of Davisville / N Appleton 138kV, the W Mount Vernon / Santa Rosa Beach 138kV loads to 112.5% of its 196 MVA rating. Project DFAX: 6.5%.

Guadalupe / Mortan 138kV: 112.4% loading (250 MVA) following the outage of the Wissota / Berry Street 138kV; DFAX = 9.2%.

For the loss of Moffat / Killen 500kV, the Moffat / Killen 230kV loads to 111.9% of its 1,010 MVA rating. Project DFAX: 25.2%.

Mainwaring Substation / Mortan 138kV: 111.9% loading (254 MVA) following the outage of the Wissota / Berry St 138kV; DFAX = 9.2%.

Wissota / Nsp 138kV: 111.4% loading (223 MVA) following the outage of the Davisville / N Appleton 138kV; DFAX = 6.5%.

For the loss of Toledo Bend / Killen 500kV Line, the Berry St 345/138kV Transformer loads to 111.3% of its 427 MVA rating. Project DFAX: 11.3%.

For the loss of Ojo / Sand Springs 138kV L, the Snohomish / Sand Springs 138kV loads to 110.8% of its 392 MVA rating. Project DFAX: 10.8%.

For the loss of Toledo Bend / Adkins 500kV line, the Toledo Bend 500/230kV Transformer Bank loads to 110.3% of its 645 MVA rating. Project DFAX: 12.8%.

The Union / Claus 138kV Line is overloaded to 110.3% of its 290 MVA emergency rating for the loss of the Toledo Bend / Killen 500kV. The project contributes 9.3% DFAX (10.0 MW) to this overload.

The W Mount Vernon / Nsp 138kV is overloaded to 110.3% of its 213 MVA emergency rating for the loss of the N Appleton / Davisville 138kV L. The project contributes 6.5% DFAX (7.0 MW) to this overload.

Sand Springs Substation / Ojo 138kV: 109.8% loading (418 MVA) following the outage of the Sand Springs / Snohomish 138kV; DFAX = 11.6%.

Leslie Substation / St Joseph 138kV: 109.8% loading (718 MVA) following the outage of the Albion Rd / Wixom Substation 138kV; DFAX = 6.7%.

N Appleton / Leslie Substation 138kV Line: 109.4% loading (726 MVA) following the outage of the Albion Rd / Wixom 138kV; DFAX = 6.7%.

For the loss of Wixom / Albion Road 138kV line, the Davisville / N Appleton 138kV loads to 109.0% of its 736 MVA rating. Project DFAX: 6.7%.

Loveland / Berry Street 138kV L: 108.1% loading (264 MVA) following the outage of the Berry St / Northfield 345kV Line; DFAX = 8.8%.

The Claus / San Adreas 138kV is overloaded to 108.0% of its 313 MVA emergency rating for the loss of the Killen Substation / Toledo Bend 500kV line. The project contributes 9.3% DFAX (10.0 MW) to this overload.

For the loss of Berry St Substation / Northfield 345kV Line, the Loveland / Ojo 138kV loads to 107.7% of its 268 MVA rating. Project DFAX: 8.8%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Douglas Forest / Ellendale 138kV L	Rebuild line with higher-capacity conductor	\$9,122,000	3% (\$253,000)
Georgia Pacific / Douglas Forest 138kV	Rebuild line with higher-capacity conductor	\$8,771,000	3% (\$243,000)
Coulee Ave / Sporn 345kV line	Rebuild line with higher-capacity conductor	\$48,193,000	1% (\$611,000)
Wellington / Davisville 138kV L	Rebuild line with higher-capacity conductor	\$5,554,000	1% (\$58,000)
Tule / Santa Rosa Beach 138kV	Rebuild line with higher-capacity conductor	\$5,048,000	3% (\$145,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Port Madison / Pinn Rd 138kV L	Rebuild line with higher-capacity conductor	\$2,755,000	4% (\$115,000)
Southside Wastewater Treatment Plant / Osawatomie 100kV	Rebuild line with higher-capacity conductor	\$2,630,000	6% (\$144,000)
Port Madison / Ojo 138kV	Rebuild line with higher-capacity conductor	\$12,263,000	4% (\$512,000)
Mistersky / Guadalupe 138kV	Rebuild line with higher-capacity conductor	\$16,558,000	3% (\$514,000)
Bingen / Prager 138kV	Rebuild line with higher-capacity conductor	\$1,289,000	3% (\$38,000)
Santa Rosa Beach / W Mount Vernon 138kV line	Rebuild line with higher-capacity conductor	\$3,366,000	3% (\$96,000)
Guadalupe / Mortan Substation 138kV	Rebuild line with higher-capacity conductor	\$13,514,000	3% (\$420,000)
Killen / Moffat Substation 230kV	Rebuild line with higher-capacity conductor	\$36,988,000	4% (\$1,398,000)
Mainwaring Substation / Mortan 138kV	Rebuild line with higher-capacity conductor	\$17,989,000	3% (\$558,000)
Wissota / Nsp 138kV	Rebuild line with higher-capacity conductor	\$25,232,000	3% (\$722,000)
Berry St 345/138kV Transformer	Replace transformer with higher-rated unit	\$14,591,000	4% (\$527,000)
Snohomish / Sand Springs 138kV	Rebuild line with higher-capacity conductor	\$22,179,000	3% (\$682,000)
Toledo Bend 500/230kV Transformer Bank	Replace transformer with higher-rated unit	\$35,355,000	4% (\$1,260,000)
Union / Claus 138kV Line	Rebuild line with higher-capacity conductor	\$2,772,000	3% (\$90,000)
W Mount Vernon / Nsp 138kV	Rebuild line with higher-capacity conductor	\$9,071,000	3% (\$260,000)
Sand Springs / Ojo 138kV	Reconductor line	\$2,963,000	3% (\$89,000)
St Joseph / Leslie 138kV	Reconductor line	\$3,956,000	1% (\$36,000)
North Appleton / Leslie 138kV	Reconductor line	\$2,451,000	1% (\$22,000)
Davisville / N Appleton 138kV	Reconductor line	\$2,151,000	1% (\$19,000)
Loveland / Berry Street 138kV L	Reconductor line	\$3,606,000	3% (\$112,000)
Claus / San Adreas 138kV	Reconductor line	\$2,986,000	3% (\$97,000)
Ojo / Loveland 138kV	Reconductor line	\$1,684,000	3% (\$52,000)
	Total allocated to project		\$9,073,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.